

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Soils: fertility, erosion & management - Set 1

### Mark scheme

AQA spec 3.2.5; The physical environment; textbook pp. 158-169

#### Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <ul style="list-style-type: none"> <li>• B (about 40:40:20 sand:silt:clay)</li> <li>• Balanced mix gives good drainage and aeration from the sand fraction</li> <li>• Water retention and nutrient (ion) retention from the clay fraction; easy root penetration and cultivation</li> </ul>                                                                                                | 3     |
| 01.2     | <ul style="list-style-type: none"> <li>• Large sand particles leave large pore spaces so water drains rapidly (high permeability / aeration)</li> <li>• Nutrient ions adsorb onto the (charged) surfaces of clay particles but not onto sand</li> <li>• With only 5% clay there is little surface to hold ions so nutrients are leached; low water retention too</li> </ul>                | 3     |
| 01.3     | <ul style="list-style-type: none"> <li>• C (clay) holds much water so has a high thermal capacity: warms and cools slowly; A warms rapidly after cold weather</li> <li>• Clay particles held together by capillary water so roots find it hard to push between them; penetration is easy in sandy A</li> </ul>                                                                             | 2     |
| 01.4     | <ul style="list-style-type: none"> <li>• Shake a known mass of soil with water in a measuring cylinder and leave to settle; particles settle in order of size (sand first, then silt, then clay; organic matter floats)</li> <li>• Measure the depth / volume of each layer to calculate the percentage of each fraction, then use the soil triangle to name the textural class</li> </ul> | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-160, 412; AQA spec 3.2.5.1.                                                                                                                                                                                                                                                               |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 02.1     | <ul style="list-style-type: none"> <li>• Detritivores (worms, beetle larvae, millipedes, woodlice) physically break up dead organic matter, increasing its surface area, and release nutrients; worms create tunnels improving drainage and aeration</li> <li>• Decomposers (bacteria, fungi) secrete enzymes that break down dead organic matter, releasing nutrient ions (ammonification) and forming humus which binds particles into crumb peds</li> <li>• Nitrogen-fixing bacteria convert nitrogen gas to ammonium ions; nitrifying bacteria oxidise ammonium to nitrite then nitrate for root absorption</li> <li>• Mycorrhizal fungi form symbiotic associations with roots increasing phosphate uptake</li> </ul> | 6     |
| 02.2     | <ul style="list-style-type: none"> <li>• This is the range of tolerance of most plants and soil biota; acidic soils leach nutrients and mobilise toxic aluminium, damaging root membranes</li> <li>• In alkaline soils phosphates become insoluble / unavailable</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2     |
| 02.3     | <ul style="list-style-type: none"> <li>• Interstitial spaces are reduced so aeration falls (aerobic processes such as nitrification and decomposition slow) and roots cannot penetrate</li> <li>• Infiltration falls so waterlogging is more likely and runoff / erosion increase</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                               | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-161; AQA spec 3.2.5.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |
| 03.1     | <ul style="list-style-type: none"> <li>• Soil loss increases as vegetation cover decreases: forest lowest (0.5), bare soil highest (60 t ha<sup>-1</sup> yr<sup>-1</sup>) - over 100 times more</li> <li>• Ploughing downslope loses nearly three times as much as contour ploughing for the same crop</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                          | 2     |
| 03.2     | <p><i>Any 4 from:</i></p> <ul style="list-style-type: none"> <li>• Leaves / leaf litter intercept raindrops so soil particles are not dislodged by rain splash</li> <li>• Roots bind soil particles together</li> <li>• Vegetation acts as a windbreak reducing wind velocity and the kinetic energy available to carry particles</li> <li>• Plants increase infiltration (root channels, organic matter) so less surface runoff</li> <li>• Organic matter / humus from vegetation binds particles into stable peds</li> </ul>                                                                                                                                                                                             | 4     |
| 03.3     | <ul style="list-style-type: none"> <li>• Furrows running downslope act as channels / gulleys that increase the velocity (kinetic energy) of runoff</li> <li>• Furrows along the contour trap the water so it loses energy, deposits particles and infiltrates</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2     |
| 03.4     | <ul style="list-style-type: none"> <li>• In the dry season the soil (especially low-clay soils) is loose and dry with little cohesion so wind blows it away</li> <li>• In the wet season heavy rain causes rain splash and surface runoff when infiltration capacity is exceeded</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |
| 04.1     | <ul style="list-style-type: none"> <li>• Overgrazing: vegetation is eaten faster than it grows so soil is exposed; hooves disturb the soil, damage roots and compact it, reducing infiltration and increasing runoff</li> <li>• Reduced biota (ploughing, agrochemicals, less organic matter): less humus so less adhesion between particles; fewer worms so less aeration and lower infiltration capacity; fewer nutrients released so less vegetation cover</li> </ul>                                                                                                                                                                                                                                                   | 4     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 04.2     | <ul style="list-style-type: none"> <li>• Tree roots that held the soil together decompose so the soil is bound less strongly</li> <li>• Heavy rain adds mass and lubricates particles; without interception more water enters the soil, making slumping more likely</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2     |
| 04.3     | <p><i>Any 4 from:</i></p> <ul style="list-style-type: none"> <li>• Sedimentation in rivers reduces flow capacity so flooding downstream is more likely</li> <li>• Sedimentation in reservoirs reduces their storage volume</li> <li>• Coastal sedimentation covering reefs / seagrass; turbidity reducing light for aquatic plants</li> <li>• Increased atmospheric particulates from wind erosion worsening asthma; dust covering crops / urban areas</li> <li>• Desertification as vegetation cannot re-establish; reduced rainfall</li> <li>• Nutrients (phosphates) carried on soil particles causing eutrophication</li> </ul>                                                                                                                                                                                                                                                                                                           | 4     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |
| 05.1     | <ul style="list-style-type: none"> <li>• <math>110 \times 0.67 \times 1.35 \times (0.50 \times 0.90) \times 1.0</math></li> <li>• <math>= 44.8 \text{ t ha}^{-1} \text{ yr}^{-1}</math> (accept 44.7-44.8)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2     |
| 05.2     | <ul style="list-style-type: none"> <li>• <math>110 \times 0.67 \times 1.35 \times (0.35 \times 0.60) \times 0.25</math></li> <li>• <math>= 5.2 \text{ t ha}^{-1} \text{ yr}^{-1}</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
| 05.3     | <ul style="list-style-type: none"> <li>• Beans / up-and-down ploughing: severe class (over 30) and about 45 times the formation rate - clearly unsustainable, soil depth will fall</li> <li>• Wheat / mulch / strip cropping: moderate class, reduced by about 88% but still 5 times the formation rate so not fully sustainable</li> <li>• Further measures (long-term crops, terracing, windbreaks, more organic matter) or grazing land (<math>C = 0.02</math>) needed to bring loss below <math>1 \text{ t ha}^{-1} \text{ yr}^{-1}</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                     | 3     |
| 05.4     | <ul style="list-style-type: none"> <li>• Factors are averages / arbitrary values for typical conditions; rainfall erosivity varies between years / storms</li> <li>• Does not include gully erosion, landslides, wind erosion or local variation in soil; based on data from particular regions</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2     |
| 05.5     | <ul style="list-style-type: none"> <li>• C (crop / tillage) or P (erosion control practice) - R, K and LS are fixed by climate, soil and topography</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 167-169; AQA spec 3.2.5.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |
| 06.1     | <ul style="list-style-type: none"> <li>• Terracing: sloping land divided into narrow level fields held by retaining walls along contours; water slows as it crosses each terrace and infiltrates instead of running off</li> <li>• Tied ridging: on nearly flat land a criss-cross of intersecting ridges retains rainwater, increasing infiltration and preventing runoff</li> <li>• Windbreaks: hedgerows / tree rows reduce wind velocity (for up to 30 times their height) so the wind has less kinetic energy to carry particles; protect bare fields after harvest</li> <li>• Mulching: organic matter on the surface protects soil from raindrop impact and wind; decomposes to humus which binds particles; retains moisture</li> <li>• Strip cropping: strips of different crops harvested at different times (at <math>90^\circ</math> to the wind) so there is always a crop protecting part of the field; slows runoff</li> </ul> | 5     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                     | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 06.2     | <ul style="list-style-type: none"> <li>• No frequent replanting / soil disturbance (ploughing) that breaks up soil structure and exposes particles</li> <li>• Permanent root systems and canopy cover protect the soil all year</li> </ul>                                                                                                                                           | 2     |
| 06.3     | <p><i>Any 3 from:</i></p> <ul style="list-style-type: none"> <li>• Control stocking density so vegetation regrows faster than it is eaten</li> <li>• Restrict access to vulnerable areas such as river banks (fencing)</li> <li>• Move / remove livestock at high-risk times, eg after heavy rain when soil is wet</li> <li>• Rotational grazing to rest pastures</li> </ul>         | 3     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 164-166; AQA spec 3.2.5.3.                                                                                                                                                                                                                                                              |       |
| 07.1     | <ul style="list-style-type: none"> <li>• Water = <math>65.0 - 55.0 = 10.0</math> g; fresh soil = 40.0 g</li> <li>• <math>10.0 / 40.0 \times 100 = 25.0\%</math></li> </ul>                                                                                                                                                                                                           | 2     |
| 07.2     | <ul style="list-style-type: none"> <li>• Organic matter = <math>55.0 - 52.6 = 2.4</math> g; dry soil = 30.0 g</li> <li>• <math>2.4 / 30.0 \times 100 = 8.0\%</math></li> </ul>                                                                                                                                                                                                       | 2     |
| 07.3     | <ul style="list-style-type: none"> <li>• So all water is removed and the mass loss on burning is due to organic matter only</li> <li>• Heating until mass no longer changes shows all water has evaporated</li> </ul>                                                                                                                                                                | 2     |
| 07.4     | <ul style="list-style-type: none"> <li>• Organic matter releases nutrients as it decomposes and provides food for soil biota</li> <li>• Increases water retention and, as humus, binds particles into crumb peds improving structure / aeration</li> <li>• Surface organic matter / humus adhesion reduce rain splash and wind erosion</li> </ul>                                    | 3     |
| 07.5     | <ul style="list-style-type: none"> <li>• Samples taken from different depths / locations (organic matter is concentrated near the surface); incomplete burning; different heating times</li> </ul>                                                                                                                                                                                   | 1     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-159, 412; AQA spec 3.2.5.1, 3.7.2.1.                                                                                                                                                                                                                                                |       |
| 08.1     | <ul style="list-style-type: none"> <li>• B - tiny pores between clay particles allow capillary rise; none in sand</li> </ul>                                                                                                                                                                                                                                                         | 1     |
| 08.2     | <ul style="list-style-type: none"> <li>• Hygroscopic: held tightly by adhesive forces on particle surfaces; cannot drain and cannot be absorbed by plants</li> <li>• Capillary: held in pores by surface tension; does not drain but can be absorbed by plants</li> <li>• Gravitational: drains freely through large pores; available to plants only briefly as it passes</li> </ul> | 3     |
| 08.3     | <ul style="list-style-type: none"> <li>• Small round crumbs leave spaces giving good drainage, aeration and easy root penetration; large flat platy peds pack together reducing all three</li> </ul>                                                                                                                                                                                 | 1     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.                                                                                                                                                                                                                                                                |       |